National University of Computer and Emerging Sciences, Lahore Campus

	Lab Name:	**Artificial Intelligence**	**Course Code:**	
	Program:	**BS (Computer Science)**	**Semester:**	**Spring 2026**
	Duration:	**15 minutes**	**Total Marks:**	**10**
	Quiz Date:	**7th May, 2026**	**Section**	**4F**
	Exam Type:	**Quiz 2**	**Page(s):**	

Student Name Roll No.

Section:

Quiz 2

Q1: True/False: (2 marks)

1. AUC value closer to 1 is better.

True : AUC closer to 1 indicates better model performance.

2. Early stopping prevents overfitting.

True : Early stopping helps prevent overfitting by stopping training when validation loss stops improving.

Q2: Fill in the missing: (1 mark)

1. K-Fold Cross Validation splits data into _______ parts.

K equal parts.

Q3: Write TensorFlow code to: **(7 marks)**

- Build an MLP with:
  - Input layer
  - 2 hidden layers (ReLU)
  - Output layer (Softmax)
- Use:
  - Adam optimizer
  - Dropout

Train for 5 epochs. Load dataset using this code:

```
import tensorflow as tf
from tensorflow.keras import layers, models

# Load dataset
(x_train, y_train), (x_test, y_test) = tf.keras.datasets.fashion_mnist.load_data()

#Mention your code on the back side of sheet
```

```
import tensorflow as tf

from tensorflow.keras import layers, models


# Load dataset

(x_train, y_train), (x_test, y_test) = tf.keras.datasets.fashion_mnist.load_data()


# Normalize data

x_train = x_train / 255.0

x_test = x_test / 255.0


# Build MLP model

model = models.Sequential([

    layers.Flatten(input_shape=(28, 28)),   # Input layer


    layers.Dense(256, activation='relu'),   # Hidden layer 1

    layers.Dropout(0.3),


    layers.Dense(128, activation='relu'),   # Hidden layer 2
```

```
    layers.Dropout(0.3),


    layers.Dense(10, activation='softmax')  # Output layer
])


# Compile model
model.compile(
    optimizer='adam',
    loss='sparse_categorical_crossentropy',
    metrics=['accuracy']
)


# Train model
model.fit(x_train, y_train, epochs=5, validation_data=(x_test, y_test))
```